Section B contains a mixture of short-answer and extended answer questions. This will include questions on the analysis and evaluation of practical work.

Students may use a calculator.

Quality of written communication will be assessed in this examination through questions which are labelled with an asterisk (*). When answering these questions students should consider spelling, punctuation and grammar of their response, as well as the clarity of expression.

1.3 Formulae, equations and amount of substance

Application of ideas from this topic will be applied to all other units.

Students will be assessed on their ability to:

a demonstrate an understanding of the terms *atom*, *element*, *ion*, *molecule*, *compound*, *empirical* and *molecular formulae*

b write balanced equations (full and ionic) for simple reactions, including the use of state symbols

c demonstrate an understanding of the terms *relative atomic mass*, *amount of substance*, *molar mass* and *parts per million (ppm)*, e.g. gases in the atmosphere, exhausts, water pollution

d calculate the amount of substance in a solution of known concentration (excluding titration calculations at this stage), e.g. use data from the concentrations of the various species in blood samples to perform calculations in mol dm^{-3}

e use chemical equations to calculate reacting masses and vice versa using the concepts of amount of substance and molar mass

f use chemical equations to calculate volumes of gases and vice versa using the concepts of amount of substance and molar volume of gases, e.g. calculation of the mass or volume of CO_2 produced by combustion of a hydrocarbon (given a molar volume for the gas)

g use chemical equations and experimental results to deduce percentage yields and atom economies in laboratory and industrial processes and understand why they are important

h demonstrate an understanding of, and be able to perform, calculations using the Avogadro constant

i analyse and evaluate the results obtained from finding a formula or confirming an equation by experiment, e.g. the reaction of lithium with water and deducing the equation from the amounts in moles of lithium and hydrogen j make a salt and calculate the percentage yield of product, e.g. preparation of a double salt (ammonium iron(II) sulfate from iron, ammonia and sulfuric acid)

k carry out and interpret the results of simple test tube reactions, such as displacements, reactions of acids, precipitations, to relate the observations to the state symbols used in equations and to practise writing full and ionic equations.